

Excipients that may be used in the formulation of suppositories

- The list of possible excipients is more limited for suppositories than for other dosage form types
- Surface-active agents;
- Agents to reduce hygroscopicity; and
- Agents to control the melting point of the base

Surface-active agents

- These are included to enhance the wetting properties of the suppository base with the rectal fluid.
- This in turn will enhance drug release/dissolution.
- The use of surfactants is mainly reserved for formulations composed of a lipophilic suppository base and/or a lipophilic drug.
- Examples of excipients, sorbitan esters and polyoxyethylene sorbitan fatty acid esters.

Agents to reduce hygroscopicity

- Water uptake during storage will result in changes to the mechanical properties (softening) and shape of these dosage forms.
- Agents that reduce hygroscopicity, e.g. colloidal silicon dioxide, may be included in fatty suppository bases to reduce the uptake of water from the atmosphere during storage

Agents to reduce hygroscopicity.....

- The hygroscopicity of water-soluble or water-miscible bases is large and is indeed partly responsible for the in vivo performance of formulations base.
- Accordingly, protection against water uptake during storage is afforded by the use of moisture-resistant packaging

Agents to control the melting point of the base

- Examples of excipients that are commonly used to increase the melting point of suppositories prepared using fatty bases include:
 - beeswax (white or yellow wax)
 - cetyl esters wax USPNF
 - stearic acid
 - atearic alcohol
 - aluminium mono- or distearate
 - colloidal silicon dioxide
 - magnesium stearate
 - bentonite.

Agents to control the melting point of the base

- Conversely there may be a requirement to reduce the melting point of the fatty suppository base, e.g. to enable melting within the rectum. Examples of excipients that may be used for this purpose include:
 - glyceryl monostearate
 - myristyl alcohol
 - polysorbate 80
 - propylene glycol.

Preparation of suppositories

Preparation of suppositories

- Suppositories are prepared by two methods:
 - ✓ hand-shaping (rolling)
 - ✓ molding (fusion)

Preparation of suppositories

❑ Hand-shaping (Rolling)

- Hand shaping supps is the oldest and simplest method.
- This manipulation **requires considerable skill**, yet avoids the complications of **heat** and **mold preparation**.

Hand-shaping (rolling).....

- This method is generally suitable for cocoa butter.
- There is little requirement for today's pharmacist to shape suppositories by hand.
- Hand rolling and shaping is a historic part of the art of pharmacist.

Hand-shaping (rolling).....

❑ Procedure

- ✓ In a mortar reduce the medicament into a **fine powder**.
- ✓ Take the correct amount of grated cocoa butter and **incorporate** the medicament with pestel.

Hand-shaping (rolling).....

- ✓ When the mass become **plastic** under the vigorous kneading of the pestel, transfer with spatula to a piece of paper kept in hand.
- ✓ By quick **rotary** movements of the hands, the mass is rolled to the ball which immediately is placed on a pile tile.

Hand-shaping (rolling).....

- ✓ A **suppository cylinder** is formed by rolling the mass on the tile with flat board.
- ✓ The cylinder is then cut into the appropriate number of pieces by spatual. Each piece resembles a supp.
- ✓ Roll one end of the supp with a spatula or with fingers to produce rounded point.

Molding (fusion)

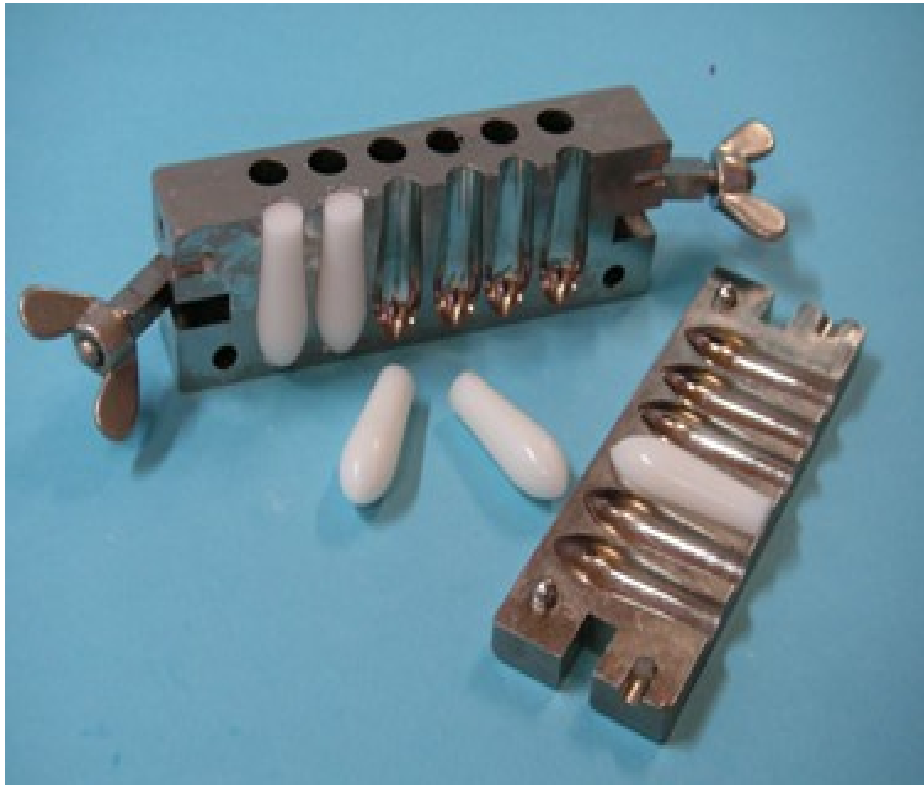
- In this method the drug is **dispersed or dissolved** in the melted supp base and poured into suppository mold.
- It is the principal method for the preparation of suppositories.

Molding (fusion)

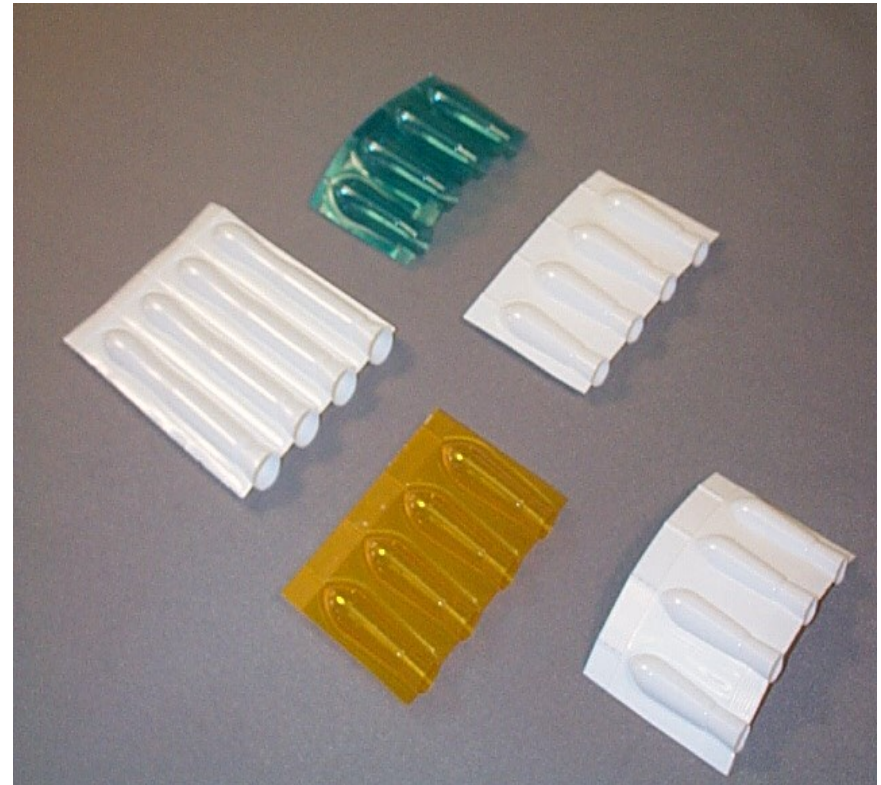
- Suppository molds are available for the preparation of varies shapes and sizes of suppository.
- Molds are made of **aluminum alloys, brass, stainless steel, plastic, or rubber** and are available with from **6** to **several hundred** cavities.

Types of molds

- Metallic mold

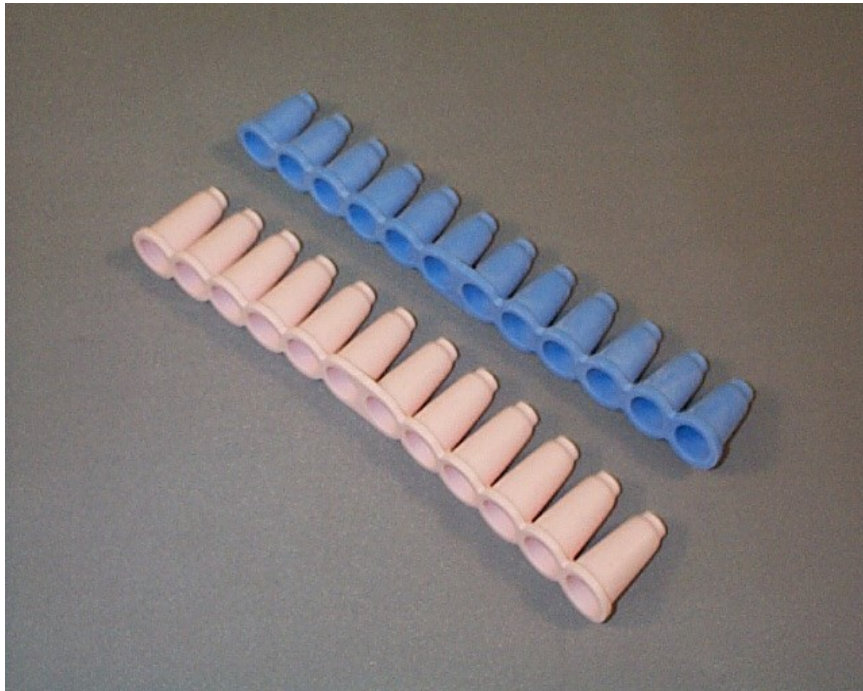


- Plastic molds

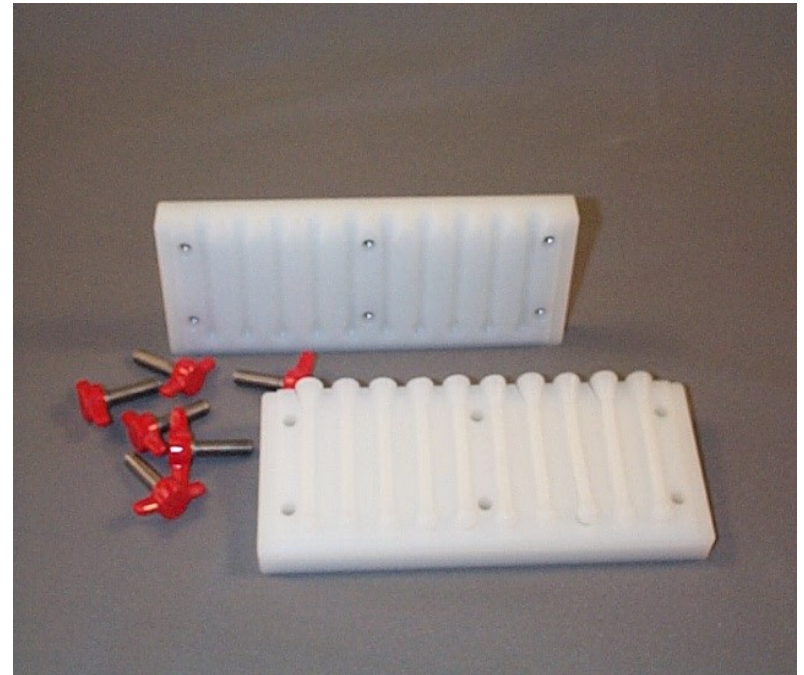


Types of molds

- Flexible rubber molds



- Very hard rubber mold



Molding (fusion).....

- **Major steps for preparation of suppositories using fusion method:**
 - ✓ calibration of the mold
 - ✓ determination of the amount of base required
 - ✓ lubrication of the mold
 - ✓ preparation of the base
 - ✓ incorporation of the drug
 - ✓ filling the mold
 - ✓ trimming (removing the excess)
 - ✓ Refrigeration
 - ✓ removing formed suppositories from mold
 - ✓ Packaging
 - ✓ Labeling

Molding (fusion).....

□ 1. Calibration of the mold

- The nominal capacities of common molds are **1 g, 2 g, 4 g**, and **8 g** when cocoa butter is used as a base.
- The nominal capacity of mold implies a calibration of mold is done using cocoa butter.
 - therefore will vary with different bases depending upon their density.

Molding (fusion).....

- Calibration of mold can be done by preparing a set of suppositories or pessaries using the base alone (blank suppositories or pessaries) weighing the products and taking the mean as the true capacity.
- Calibration of mold is not needed for semi-synthetic-fats as these bases are formulated to density equal to cocoa butter.

Molding (fusion).....

○ 2. Determination of the amount of bases required.

- ✓ this can be determined by using displacement value (density factor) of drugs in a suppository base.
- Displacement value (density factor) of a drug in a suppository base can be obtained from reference sources (e.g., books) or can be determined in lab.

Molding (fusion).....

- **Displacement value (density factor):** *defined as the weight of the medicament in gram that displaces 1 g of the base.*

$$DV_{drug} = \frac{\text{Weight of drug to be added}}{\text{Weight of base to be displaced (removed)}}$$

or

$$DV_{drug} = \frac{\text{density of drug}}{\text{density of base}}$$

Molding (fusion).....

- ✓ For example, Zinc Oxide is five times denser than cocoa butter

$$DV = 5 = \frac{\text{weight of drug to be added}}{\text{weight of base to be displaced (removed)}}$$

$$\rightarrow \text{Wt. of base to be displaced} = \frac{\text{Weight of drug to be added}}{5}$$

Molding (fusion).....

- If we want to add 200 mg of ZnO in 1 g mold, part of the base should be displaced

$$\text{weight of base} = \frac{200 \text{ mg}}{5} = 40 \text{ mg}$$

- Therefore, weight of base required to prepare a single supp = $1000 \text{ mg} - 40 \text{ mg} = 960 \text{ mg}$.

Molding (fusion).....

- **Displacement values of some drugs in cocoa butter:**

Aspirin.....1.3

Bismuth subgallate.....2.7

Chloral hydrate.....4.5

Morphine HCl.....1.6

Phenobarbitone.....1.2

Molding (fusion).....

- To determine the displacement value of a drug in laboratory for a particular base (**double-casting method**) :
 - ✓ Prepare blank supps and take the mean weight
 - ✓ Prepare medicated supps each having 'X' amount of drug (v/v) and take the mean weight
 - ✓ Calculate the mass of the base and drug(w/w)
 - ✓ Calculate the mass of base displaced by X' amount of drug
 - ✓ Calculate the displacement value for the drug.

Molding (fusion).....

- Example, determine the displacement value for a therapeutic agent in a triglyceride base. The concentration of drug in each suppository should be 10%.

Step 1

- Prepare (e.g. six) blank suppositories and weigh. Total weight determined as 6 grams (1 gram per suppository).

Molding (fusion).....

Step 2

- Prepare six suppositories containing 10% drug in each suppository (based on the volume of the suppository mould) and weigh.
- Total weight determined as 6.11 grams.

Step 3

- Calculate the masses of base and drug in the six suppositories (remembering that the drug loading of the suppositories is 10%).

Molding (fusion).....

$$\text{Base} = 90/100 \times 6.11 = 5.50g$$

$$\text{Drug} = 10/100 \times 6.11 = 0.61g$$

Step 4

- Calculate the mass of base displaced by the 0.61 gram of drug

$$6.0 - 5.5 = 0.5g$$

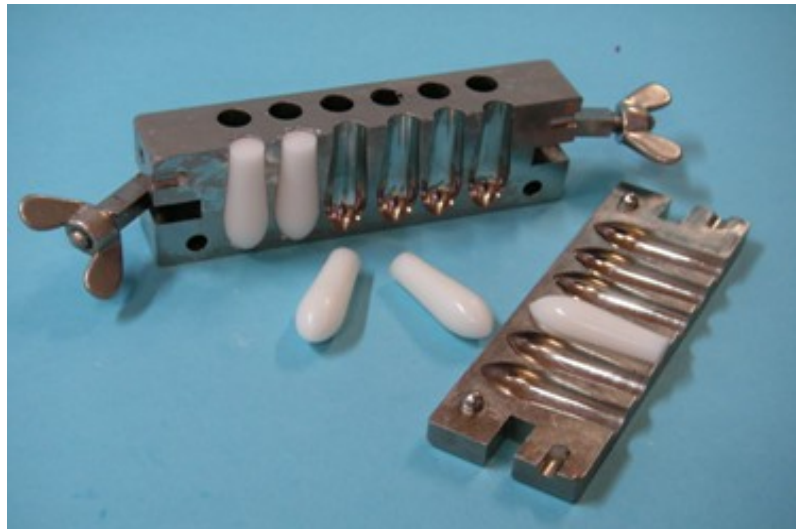
Step 5

- Calculate the displacement value for the drug.
- $DV = 0.61g / 0.5g = 1.22$

Molding (fusion).....

3. Lubrication of the molds

- Lubrication is necessary for cocoa and glycerogelatin bases because they do not contract significantly upon cooling.



Molding (fusion).....

4. Preparation of bases

Preparation of fatty bases

- ✓ Weigh shredded base in porcelain dish and warm gently using a water bath or hot plate.
- ✓ Allow two-third of the base to melt and remove from the heat.
- ✓ stir gently until the residual mass melts

Molding (fusion).....

Preparation of glycerol-gelatin base

- ✓ Weigh and then heat glycerin using suitable dish to 100 °C.
- ✓ Weigh water into a second tared dish and heat until boiling. Then remove from the heat source to prevent excess evaporation.

Molding (fusion).....

- ✓ Add gelatin to the hot water and stir gently to wet the material thoroughly.
- ✓ Add the hot glycerin to the dish containing gelatin solution and place it on a boiling water bath until dissolution is complete.

Preparation of glycono-gelatin base

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- ✓ When dissolution is complete, weigh the dish and contents and, if necessary, evaporate or add hot purified water to obtain the correct weight.
- After adjustment of weight any large air bubbles can be removed by passing the molten mass through a warm muslin strainer into a second hot dish.

Molding (fusion).....

Preparation of PEG bases

- These are prepared the same way as fatty bases with special attention to the following:
- ✓ the constituent PEGs are melted together
- ✓ If the base contains water, this should be warmed to the same temperature, added to the melted base and the mixture is stirred.

Molding (fusion).....

5. Incorporation of the medicament

- Insoluble medicaments should be powdered if necessary and sieved before weighing.
- Other ingredients should be weighed or measured.

Molding (fusion).....

Incorporation of the medicament in fatty bases

- ✓ **Soluble** and **heat stable ingredients** are incorporated directly into the molten base and stirred to form a homogenous mixture.
- ✓ For **insoluble** and **poorly dispersible ingredients**, add about half of the molten base to the powdered drug and rub together in ointment tile or mortar and pestle.

Molding (fusion).....

- ✓ If the base solidifies, it can be softened by holding the tile over the water bath or hot plate for few seconds.
- ✓ Transfer the dispersion to the dish and stir to form a homogenous mixture.
- ✓ The homogenous mixture in the dish should be stirred until the base begins to thicken.

Molding (fusion.....)

- **Incorporation of the medicament in glycerogelatin bases**
 - ✓ Soluble and heat stable medicaments are dissolved in a little water before adding to molten mass.
 - ✓ Insoluble medicaments are rubbed down with part of glycerol.

Molding (fusion).....

- **Incorporation of medicaments in PEG bases**

- ✓ If water is component of the base, water soluble and thermostable medicaments are dissolved in water before mixing with molten PEG.
- ✓ Insoluble medicaments are incorporated in the same method as described in fatty bases.

Molding (fusion).....

6.Filling the mold

- ✓ The mold is filled when the base starts **thickening**.
- ✓ For semi-synthetic fat and PEG bases, fill each cavity in the mold excessively to allow contraction on cooling.
- ✓ If the consistency of the base is correct the overflow should remain above each cavity.

Molding (fusion).....

7.Trimming (removing the excess)

- ✓ Leave the mass **to set/cool** for 2-3 minutes for contraction of the base to occur.
- ✓ Remove the excess mass by blade of firm spatula.
- ✓ Place the blade of the spatula at one end of the mold and draw it across the mold pressing down at the same time.

Molding (fusion).....

8.Refrigerating

- ✓ In order **to hasten hardening** of suppositories, they can be placed in refrigerator for short time (e.g., 10 minutes).

9.Open the mold

- ✓ After 10 minutes standing in a cool place, the mold may be allowed to equilibrate to room temperature.
- ✓ Finally, the supps are removed from the mold, inspected and packed.

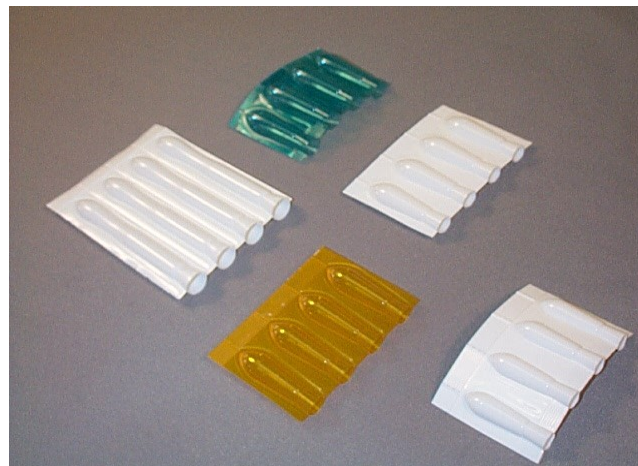
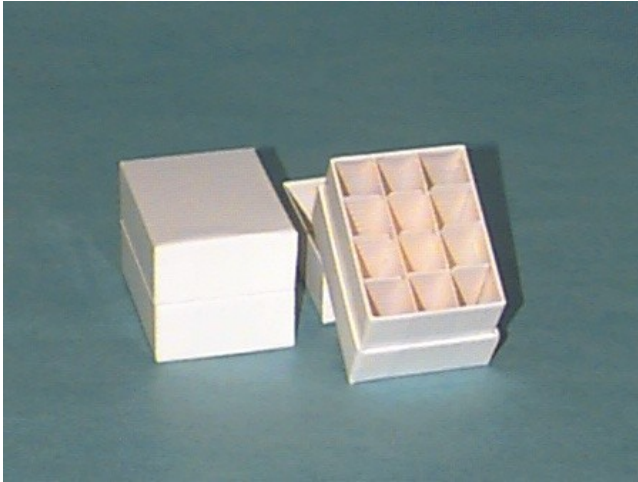
Packaging

- Supps must be packed or wrapped individually or separated in compartmented boxes so that they do not adhere each other.
- All PEG based supps should not be stored in plastic containers.

Packaging

- For hygroscopic suppositories **tightly sealed glass or plastic containers** should be used.
 - Suppositories stored at high humidity may absorb moisture and tend to become spongy
- For light sensitive drugs, metallic foils or amber coloured containers are used to wrap each supps. example, Al foils, tin foils

Packaging....



Labeling information for extemporaneously prepared suppositories

- ✓ Name and address of the pharmacy
- ✓ Name of the suppository and its weight
- ✓ Strength of the medicament. (e.g., paracetamol 125 mg)
- ✓ Quantity and list of ingredients used if the formula used is not from pharmacopeia
- ✓ Date of preparation
- ✓ Beyond use date/ expiry date

Labeling requirement for extemporaneously prepared suppositories

- ✓ Storage conditions
- ✓ Name of the compounding pharmacist
- ✓ Name of the patient
- ✓ Precaution/Supportive information

e.g., “ for rectal use only” or

“ for vaginal use only”

“Do not swallow”

Storage

- ✓ Cocoa butter and hydrogenated vegetable oil suppositories are stored in refrigerator (2-8 °C).
- ✓ Glycerinated gelatin and PEG suppositories can be stored at room temperature (20-25 °C).
- Suppositories stored in places of extreme dryness may lose moisture and become brittle.

Advantages and disadvantages of rectal dosage forms

Advantages

- Rectal dosage forms may be successfully employed to provide a local effect for the treatment of infection and inflammation, e.g. haemorrhoids, proctitis.
- Rectal dosage forms are used to promote evacuation of the bowel (by irritating the rectum), to relieve constipation or to cleanse the bowel prior to surgery.

Advantages.....

- To provide systemic drug absorption in situations where oral drug absorption is not recommended.
- Patients who are unconscious, vomiting
- Drugs that are prone to degradation in the stomach
- Drugs that are erratically absorbed from the upper gastrointestinal tract
- Administration of drugs that are extensively first-pass metabolized.

Disadvantages

- Special advice is required concerning the administration of dosage forms.
- The absorption of therapeutic agents from the rectum is slow and prone to large intrasubject and intersubject variability
- The presence of faeces within the rectum considerably affects both the rate and extent of drug absorption.

Disadvantages

- Rectal administration of therapeutic agents may result in the development of local side-effects, in particular proctitis.
- The industrial manufacture of suppositories is more difficult than for other common dosage forms.

Factors affecting the rectal absorption of drugs

- The process by which drugs are absorbed into the systemic circulation involves
- the dissolution of the drug in the rectal fluids
- often preceded by the dissolution/melting of the dosage form), diffusion of the drug through the rectal fluids to the rectal mucosa and then absorption.
- There are various factors that affect the rate and extent of drug absorption,

Factors affecting.....

Site of absorption with in the rectum

- the site of absorption will affect the fate of the therapeutic agent within the blood stream.
- it is difficult to ensure that drug absorption occurs exclusively via the inferior and middle haemorrhoidal veins and, therefore, first-pass metabolism of therapeutic agents will occur following rectal administration
- It is markedly lower when compared to drug absorption following oral administration

Factors affecting.....

The partition coefficient and degree of ionization of the drug

- Some rectal base are **lipophilic and**, if the **therapeutic agent is also lipophilic**, the **release** of the drug will **be slow** and the **solubility** of the drug in the rectal fluids will be low
- Conversely, the use of (dispersed) hydrophilic drugs in lipophilic bases is preferred, as these would exhibit a greater thermodynamic tendency to dissolve in the rectal fluid.

Factors affecting.....

- the **limited volume of fluid in the rectum** affects both the **rate and extent** of drug dissolution therefore, **increasing the drug loading** in suppositories **does not enhance** drug absorption.
- **Degree of ionisation** of therapeutic agents affects the resultant absorption by biological membranes, as absorption requires the presence of the **unionised form** of the drug.

Factors affecting.....

The particle size of the dispersed active agent

- The rate of dissolution of the drug is inversely proportional to the particle size of the dispersed active agent.

The physicochemical properties of the formulation base

- the formulation should be is **non-irritant** to the **rectal mucosa as irritation** frequently results in **defecation** and hence removal of the dosage form before the required mass of drug has been released.
- **Hydrophobicity/lipophilicity**

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